

HSS201 History of Science
First Mid-Semester Examination

Total 25 Marks

1:00 hr

Section A (Answer Any 13 from the following): 13 Marks

- ✓ 1. The ideas of Enlightenment proclaimed which among the following human qualities to throw off ancient superstition and provide a better foundation for society?
 - (a) Religious Belief
 - (b) Happiness
 - (c) Faith in God
 - (d) Reason

- ✓ 2. According to William Whewell's *History of Inductive Sciences*, the scientific approach consisted of -
 - (a) Rigorous testing of new hypotheses by observation and experimentation
 - (b) Validating the ideas contained in scriptures
 - (c) Following the ideas of natural philosophers
 - (d) Discovering the hidden secrets of nature

- ✓ 3. The development in which of the following domains formed the context for Draper's *History of the Conflict between Science and Religion* during the 19th century?
 - (a) Chemical Revolution
 - (b) Copernican Revolution
 - (c) Darwinian Revolution
 - (d) French Revolution

- ✓ 4. What is the argument of the reactionary histories of science like Alfred North Whitehead's *Science and the Modern World* written in response to Enlightenment ideas?
 - (a) That mathematics is a language of nature
 - (b) That the working of nature can be explained rationally
 - (c) That nature is a handiwork of the God
 - (d) That there are hidden laws in the nature

- ✓ 5. By presenting the class character of the functioning of modern science and the science-society relationship, J.D. Bernal, in his works such as *Social Function of Science* and *Science in History*, adopted one of the following methods in writing the history of science?
 - (a) Historical Materialism
 - (b) Dialectical Idealism
 - (c) Idealist Relativism
 - (d) Relativist Functionalism

- ✓ 6. In 1931, Boris Hessen presented an influential paper on Newton's *Principia* in which he foregrounded externalist causes to make sense of Newton's intervention. Identify the appropriate sentence from the following that suit Hessen's thesis:
 - (a) Newton was a great scientist whose work *Principia* created a revolution in science
 - (b) Newton's *Principia* demonstrated that mathematics is the language of nature

- (c) Newton's *Principia* exhibits the complete coincidence of the physical thematics of the period, which arose out of the need of economics and technique
- (d) Newton's *Principia* contains crucial evidence of the formulas from earlier natural philosophers

7. One of the serious limitations of the histories of science written during the first half of the 20th century by such historians as Herbert Butterfield, Rupert Hall, Alexander Koyre and others was:

- (a) Great Man Theory of History of Science
- (b) Eurocentrism
- (c) Ethnonationalism
- (d) Gendered History of Science

8. What are the two themes that emerged in the history of science during the inter-war years leading to the decolonization of large parts of Africa and Asia

- (a) Globalization of modern science and comparative studies
- (b) Science in the periphery and Capitalism
- (c) Science in the metropolitan centre and Imperialism
- (d) Colonialism and Capitalism

9. In his preface to the second edition of the *Critique of Pure Reason*, Immanuel Kant used the word revolution to refer to

- (a) Scientific exploration by the Europeans in the colony
- (b) Development of Heliocentric Theory of the Universe by Copernicus
- (c) The emergence of experimental method and laboratory from the time of Galileo
- (d) Discovery of gravity by Newton

10. An approach that aimed to connect sciences and the humanities in the history of science spearheaded by Belgian-born American chemist and the historian of science George Sarton is known as

- (a) Radical Humanism
- (b) Historical Materialism
- (c) New Humanism
- (d) Nationalist Scientism

11. According to Kuhn, the period of crisis in science involves

- (a) relative peace with community of scientists working together
- (b) anarchy among the community of scientists preventing scientific activity
- (c) extraordinary research with a proliferation of competing articulations
- (d) the emergence of extraordinary scientists and philosophers

12. In Kuhnian theory of history of science, revolutions in science are

- (a) Cumulative of facts and theory
- (b) Disruptive of established theory
- (c) Continuity of theory
- (d) Erasure of a possibility of new theory

13. The influential paper of George Basalla inspired historians of science to launch:

- (a) History and Philosophy of Science in the West
- (b) Science and Empire Studies

- (c) Metropolitan Histories of Science
- (d) History of the Scientific Revolution in the West

14. Which among the following do not belong to Normal Science

- (a) Matching of facts with theory
- (b) Discarding old theories and formulating theories
- (c) Determination of significant facts
- (d) Articulation of theory

15. J.D. Bernal's *Science in History* (1954) was a monumental attempt to depict science as a potential force for good but then caught in the vortex of

- (a) Feudal interest
- (b) Mercantilist complex
- (c) Theological complex
- (d) Military-industrial complex

16. According to Kuhn, a paradigm should be

- (a) Sufficiently unprecedented to attract an enduring group of adherents
- (b) Invalidated by facts
- (c) Falsified by the community of scientists
- (d) Must have an eternal truth about the universe

Section B

Answer Any 3 from the following: 12 Marks

1. What is Needham's question in History of Science? Elaborate.
2. Define the following Kuhnian concepts: Anomaly, Incommensurability
3. What are the limits of George Basalla's three stage model in the story of spread of modern science?
4. J.D. Bernal's writings explained the class character of science in history. Elaborate.
5. What were the objectives of the UNESCO project *Scientific and Cultural History of Mankind* spearheaded by Joseph Needham?